Enrichment Analysis Results — All Models

Modality labels vs. hallucination-driving neurons

MH-Neuron / Modality Taxonomy (step 10, halluc_score)

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Scope and status

One section per model. Sections are regenerated from the enrichment JSONs on disk by `build_enrichment_pdf.py`; re-running it picks up new combos as their cluster jobs finish. `qwen2.5-vl-3b` is complete; the other models are filled in incrementally and any combo not yet on disk is marked pending.

What the numbers mean. For each modality category (visual / text / multimodal / unknown) we ablate the top 5% of neurons by ΔHallucination and run Fisher's exact test on whether that category is over-represented among those hallucination-driving neurons. The *odds ratio* (OR) is the headline: OR > 1 = enriched, OR < 1 = depleted. Two tasks are scored: **POPE** (visual object hallucination) and **TriviaQA** (text knowledge errors).

1 Model: `qwen2.5-vl-3b`

Complete — all three label/hook combos. All combos: top-$k = 5\%$, $n_{\text{driving}} = 19{,}814$ of 396,288 total neurons.

1.1 POPE (visual task) — odds ratios

Hypothesis: *visual* neurons should be enriched (ablating them blinds the model). p-values in parentheses; ≈0 denotes $p < 10^{-300}$.

category	PMBT · gate	PMBT · gate_up	FT · gate
visual	**2.02** (≈0)	**1.38** (7e-106)	1.18 (5e-25)
text	0.81 (3e-38)	0.91 (4e-9)	0.78 (1e-35)
multimodal	1.04 (0.09, n.s.)	0.93 (8e-4)	0.93 (2e-5)
unknown	**0.03** (≈0)	**0.02** (≈0)	1.06 (8e-5)

1.2 TriviaQA (text task) — odds ratios

Hypothesis: *text* neurons should be enriched (ablating them erases knowledge). These were computed from the merged ΔH_{TQA} scores (see Caveats).

category	PMBT · gate	PMBT · gate_up	FT · gate
visual	0.88 (3e-19)	1.40 (1e-115)	1.48 (7e-140)
text	**1.21** (1e-36)	0.86 (7e-23)	1.05 (0.01)
multimodal	0.85 (8e-11)	0.99 (0.6, n.s.)	1.10 (1e-7)
unknown	1.03 (0.2, n.s.)	0.08 (≈0)	0.58 (1e-232)

1.3 The double dissociation (the key result)

The dual-source design predicts a diagonal flip: visual neurons enriched on POPE but *not* TriviaQA, text neurons enriched on TriviaQA but *not* POPE. ↑ = enriched (OR>1), ↓ = depleted (OR<1).

	visual OR		**text** OR	
combo	POPE	TQA	POPE	TQA
PMBT · gate	2.02 ↑	0.88 ↓	0.81 ↓	1.21 ↑
PMBT · gate_up	1.38 ↑	1.40 ↑	0.91 ↓	0.86 ↓
FT · gate	1.18 ↑	1.48 ↑	0.78 ↓	1.05 ↑

Only PMBT · gate shows the clean flip (visual ↑↓, text ↓↑). The other two do not: under PMBT gate_up and FT gate, visual neurons are enriched for *both* tasks, so the modality axis is not separated.

1.4 Interpretation

Taxonomy: PMBT ≫ FT (same gate hook, POPE). PMBT concentrates visual function far harder (OR **2.02** vs FT's 1.18) and **purges the "unknown" bucket to 0.03**, whereas FT leaves unknown slightly *enriched* at **1.06**. So FT's catch-all category still hides hallucination-driving neurons that PMBT correctly reclassifies as visual. This is the functional-validity argument for the permutation-test labels.

Hook: gate > gate_up for separation. Both PMBT hooks share the qualitative signature (visual enriched, unknown purged ≈0.02–0.03), confirming the effect under an *independent* labeling. But the `gate` hook gives sharper visual enrichment (2.02 vs 1.38) *and* is the only one that yields the double dissociation; the `gate_up` intermediate captures visual function but does not separate the text axis (visual enriched on TriviaQA too, OR 1.40).

Headline. On identical neurons, ablation scores, and statistics, **PMBT labels at the gate hook are the only configuration that (i) maximises visual enrichment, (ii) empties the "unknown" bucket, and (iii) produces the visual/text double dissociation across POPE and TriviaQA.** FT labels achieve none of these cleanly.

1.5 Caveats

- **TriviaQA enrichment was computed offline.** The 10C aggregation historically wrote only the POPE `enrichment_results.json`; the merged ΔH_{TQA} existed (`ablation_scores_tqa.json`) but was not run through enrichment. The TQA numbers here were produced by feeding that file to the same `compute_enrichment` routine. A pipeline fix (forwarding the TQA scores so `enrichment_results_tqa.json` is emitted automatically) has been applied for the remaining models.
- Multimodal results are mostly non-significant or weak and are not load-bearing.
- "gate" and "gate_up" label *different* neuron populations (different hook point), so cross-hook agreement is corroboration by an independent labeling, not a same-neuron comparison.

2 Model: `internvl2.5-8b`

Combos available: PMBT · gate: POPE=y, TQA=y; PMBT · gate_up: POPE=y, TQA=y; FT · gate: POPE=–, TQA=–.

top-k = 5%, $n_{\mathrm{driving}} = 22,937$ of 458,752 total neurons (first available combo).

2.1 POPE (visual task) — odds ratios

Hypothesis: *visual* neurons enriched. p in parentheses; ≈0 denotes $p < 10^{-300}$.

category	PMBT · gate	PMBT · gate_up
visual	0.48 (≈0)	**2.11** (≈0)
text	**1.81** (≈0)	0.62 (3e-246)
multimodal	0.98 (0.3, n.s.)	0.71 (8e-98)
unknown	**22.82** (2e-15)	**0.00** (2e-18)

2.2 TriviaQA (text task) — odds ratios

Hypothesis: *text* neurons enriched.

category	PMBT · gate	PMBT · gate_up
visual	**1.52** (5e-204)	1.46 (5e-159)
text	0.74 (9e-106)	0.82 (6e-44)
multimodal	0.83 (8e-24)	0.82 (3e-37)
unknown	0.00 (0.4, n.s.)	0.21 (3e-09)

2.3 Double dissociation

↑ = enriched (OR>1), ↓ = depleted (OR<1). Only combos with both tasks scored are shown.

	visual OR		**text** OR	
combo	POPE	TQA	POPE	TQA
PMBT · gate	0.48 ↓	1.52 ↑	1.81 ↑	0.74 ↓
PMBT · gate_up	2.11 ↑	1.46 ↑	0.62 ↓	0.82 ↓

3 Model: `llava-onevision-7b`

Combos available: PMBT · gate: POPE=–, TQA=–; PMBT · gate_up: POPE=–, TQA=–; FT · gate: POPE=–, TQA=–.

No enrichment results on disk yet — layers still running.

4 Model: `llava-1.5-7b`

Combos available: PMBT · gate: POPE=–, TQA=–; PMBT · gate_up: POPE=–, TQA=–; FT · gate: POPE=–, TQA=–.

No enrichment results on disk yet — layers still running.